

Steering Policy Optimization with Sparse Counterfactual Advantages

Paper pitch and writing blueprint

Abstract

Policy-gradient agents are usually controlled through rewards, although the policy is updated through an advantage estimator. We study *counterfactual advantage steering*: augmenting PPO’s actor update with a policy-centered, action-selective counterfactual advantage while leaving the environment reward, GAE targets, and critic update unchanged. We first show that a state-only advantage perturbation cancels from the expected policy gradient; behavioral control therefore requires an action-selective residual. Guided by this observation, our method queries exact local counterfactual action values at a small subset of rollout states and injects the resulting centered landscape into PPO’s sampled-action objective. Across long-horizon visual control tasks, sparse counterfactual queries consistently improve learning relative to PPO and often match or exceed a more expensive direct counterfactual baseline. In contrast, globally propagating sparse labels through an imperfect distilled landscape can be unreliable. These results identify sparse, action-aligned counterfactual advantages as a lightweight control interface for policy optimization.

Abstract 2: problem–need–method–importance

Abstract

PPO is typically guided by modifying rewards, but reward design is often expensive, changes the task being optimized, and can unintentionally alter the critic’s learning target. This is especially limiting in structured simulators, where a small number of local counterfactual evaluations may reveal which action is preferable without requiring a new reward function. We ask whether this privileged but sparse information can be used to guide the policy update directly. We introduce *counterfactual advantage steering*, which adds a policy-centered, action-selective counterfactual advantage to PPO’s actor loss at a small subset of rollout states, while preserving the environment reward, GAE returns, and critic update. Our analysis shows why action selectivity is necessary: state-only advantage offsets cancel from the expected policy gradient. Empirically, sparse exact counterfactual queries improve learning on long-horizon visual control tasks and are often more reliable than applying a distilled counterfactual estimate everywhere. By separating policy guidance from reward and value-target design, this work provides a practical route for using structured simulators to make policy optimization more efficient and controllable.

One-sentence paper claim

A small number of exact, policy-centered counterfactual advantages can steer PPO’s actor update more effectively than either ordinary PPO or a noisy dense approximation, without modifying rewards or critic targets.

Scope. This is deliberately a paper about *sparse exact counterfactual advantage queries*, not about TD clipping, generic advantage bias, a learned dense landscape, or reward shaping. The secondary-goal work supports the intuition that the actor advantage is a control interface, but it is not a primary empirical claim in this version of the paper.

1 Introduction: why this problem matters

Opening. Start with the mismatch between how PPO is commonly controlled and how it is actually optimized. Rewards and critic targets are slow, global control mechanisms. The actor, however, only sees a scalar advantage associated with the sampled action. A simulator or structured environment can sometimes answer the local question “what would happen under the other actions at this state?” The question is whether a small number of such local answers can improve policy optimization without rewriting the reward function.

Gap. Full counterfactual PPO is computationally expensive because it needs an all-action landscape at many states. An obvious alternative is to learn a landscape from sparse labels and apply it everywhere. The experiments show that this alternative is not automatically safe: approximation error in the action-relevant direction can hurt the policy even when the student is trained on the same label budget.

Contribution paragraph to write. State exactly three contributions.

1. We isolate the action-selective component of an advantage perturbation as the part that affects the expected actor gradient.
2. We introduce PPO-SCA: PPO with sparse exact counterfactual advantage queries, injected into the ordinary sampled-action actor loss while rewards, returns, and critic targets are untouched.
3. We show on four main MiniGrid tasks that sparse exact queries improve learning over PPO, compare favorably with direct PPO-CF, and are more reliable than naively applying an imperfect distilled landscape everywhere.

Do not write in the introduction. Do not lead with TD clipping, adversarial attacks, backdoors, uncertainty selection, active distillation, or the custom secondary-goal environment. Those are either separate projects or ablations which obscure the central algorithmic claim.

2 Background and mechanism: advantage perturbations

Core derivation. Let an actor use an altered estimator

$$\hat{A}^f(s, a) = \hat{A}(s, a) + f(s, a). \quad (1)$$

The additional expected policy-gradient contribution is determined by

$$\Delta_f(s, a) = f(s, a) - \mathbb{E}_{a' \sim \pi(\cdot|s)}[f(s, a')]. \quad (2)$$

This equation should be the central conceptual figure or boxed proposition: the state-only portion of f is removed by the score-function identity, while the policy-centered action-dependent residual remains.

Result to use from the advantage-perturbation folder. Use the *mechanistic control*, not TD clipping. The relevant experiment is the matched state-only versus action-selective perturbation result from `Adv_perturbation_experiments/TD_CLIPPING.zip`. It establishes the following qualitative fact: at the same realized perturbation scale, a state-varying but action-independent term is an exact null for the expected actor gradient, whereas an action-selective term changes learning. Present it as a sanity check for Equation (2), not as a second main benchmark suite.

Suggested figure.

Figure 1: Action selectivity is necessary. Plot performance or update displacement against perturbation strength for clean, state-only ($\beta = 0$), and action-selective ($\beta = 1$) perturbations. Hold realized perturbation scale fixed. The visual message is “same magnitude, different effect.”

3 Method: sparse counterfactual advantage queries

Counterfactual landscape. At a queried simulator state s , evaluate local action values $Q_{\text{CF}}(s, a)$ and policy-center them:

$$A_{\text{CF}}(s, a) = Q_{\text{CF}}(s, a) - \sum_{a'} \pi_{\text{old}}(a' | s) Q_{\text{CF}}(s, a'). \quad (3)$$

The centering is essential: it makes the intervention explicitly action-selective and prevents a state-wide offset from being mistaken for useful policy guidance.

PPO-SCA update. For a queried transition, modify only the sampled-action actor advantage:

$$A_{\text{actor},t} = \text{norm}(A_{\text{GAE},t}) + \beta \frac{A_{\text{CF}}(s_t, a_t)}{\sigma_{\text{GAE}}}. \quad (4)$$

For unqueried transitions, use standard normalized GAE. In prose, explicitly state the three invariants:

1. The environment reward is unchanged.
2. GAE returns and value targets are unchanged.
3. The counterfactual term is detached and affects only the actor loss.

This distinguishes the method from reward shaping, critic-target shaping, and an auxiliary value function.

Query budget. The main setting uses exact all-action counterfactual landscapes at 2% of rollout states and uses $\beta = 0.75$. Before submission, write down a fixed selection rule for β (or retain the full $\{0.25, 0.75, 1.5\}$ sweep in the appendix). The paper must not imply that $\beta = 0.75$ was uniquely selected after inspecting only the best curves.

Pseudocode. Include a short algorithm box: collect PPO rollout; uniformly choose 2% of states; restore each selected simulator state and evaluate actions; form Equation (3); add the detached sampled-action term in Equation (4); run ordinary PPO update. Keep this under half a page.

4 Experimental protocol

4.1 Algorithms retained in the main paper

Keep only the following three learning curves in the main figures.

Method	Purpose
PPO (GAE)	Standard policy-optimization baseline.
Direct PPO-CF	Counterfactual baseline using its direct all-action objective; shows what sparse queries are competing with.
PPO-SCA (2% exact queries)	Proposed method: exact centered landscape added to the sampled-action actor advantage.

Methods removed from the main comparison. Remove the following from the title, abstract, main figures, and headline table: uniform distillation, uncertainty distillation, active distillation, multiple student-selection schemes, TD clipping, shuffled/attack variants, and manual secondary-goal steering. They create an unreadable six-method plot and weaken the paper’s causal message.

Where the removed methods go. Put uniform distillation in an ablation subsection because it motivates using exact sparse labels directly. Put active/uncertainty selection in the appendix as preliminary evidence only; it currently has three seeds. Put the secondary-goal result in the appendix or a final qualitative discussion. Do not silently delete negative results: show all removed variants in an appendix table with their configurations and seeds.

4.2 Environments and selection

The strongest four-panel learning-curve figure from `notebooks/06_plot_runs_eval_training_curves.ipynb` is:

1. MiniGrid-RedBlueDoors-6x6-v0;
2. MiniGrid-KeyCorridorS3R3-v0;
3. MiniGrid-DoorKey-6x6-v0;
4. MiniGrid-UnlockPickup-v0.

KeyCorridor S3R2 should be in the appendix because its outcomes are visibly more variable. The pre-specified E2/E3 panel also includes Taxi. Re-export Taxi in the same evaluation format before submission so that the formal four-task table can use Taxi, DoorKey, UnlockPickup, and RedBlueDoors without mixing evaluation pipelines.

Selection disclosure. The current four visual environments were selected after inspecting existing curves. Do not make a post-selected average the primary inferential result. Either (i) freeze the proposed four-task set and run fresh confirmatory seeds, or (ii) report every completed task, including KeyCorridor S3R2, in the main aggregate and reserve the four-panel figure for readability. This is important for an ICLR submission.

4.3 Evaluation and reporting

Report mean success rate and episodic return as training curves with five-seed mean and uncertainty bands. Use time-to-threshold (e.g., frames to 90% success) and final evaluation success in the main table. A normalized area-under-curve statistic can be reported in the table or appendix as a summary, but it should not be the centerpiece of the abstract.

5 Results

5.1 Main result: sparse exact counterfactual queries improve learning

What to write. Open this section with the qualitative conclusion, then point to the curves: “Across the displayed tasks, sparse exact counterfactual queries improve or accelerate PPO learning. The largest gains occur in tasks where the policy must make action-critical sequential decisions, rather than merely discover a terminal reward.” Avoid saying the method universally dominates direct PPO-CF; DoorKey is an explicit counterexample.

Environment	Normalized success during training			Frames to 90% success
	PPO	Direct PPO-CF	PPO-SCA	PPO $\rightarrow$ PPO-SCA
RedBlueDoors 6x6	0.528	0.636	0.710	848k $\rightarrow$ 695k
KeyCorridor S3R3	0.677	0.655	0.826	268k $\rightarrow$ 152k
DoorKey 6x6	0.492	0.633	0.534	285k $\rightarrow$ 273k
UnlockPickup	0.588	0.656	0.754	653k $\rightarrow$ 404k

Table note to include in the final paper. The normalized-success column above is a working extraction from the five-seed curves in Notebook 06, not yet a locked camera-ready statistic. Recompute it in one evaluation script, record uncertainty intervals, and state the exact integration convention. The numbers are useful for selecting the story but must be reproducible before being treated as final paper values.

Figure 2. Use a four-panel success-rate learning curve: PPO in blue, direct PPO-CF in orange, and PPO-SCA in green. Do not include active distillation. The existing focused plot can be simplified by removing its black active-distilled line.

5.2 Sparse exact labels are safer than naive dense distillation

What to write. This is the key ablation, not a co-equal method. The question is whether the same 2% label budget should be used immediately at queried states or learned and broadcast to all states. Write: “The benefit is not explained simply by adding a dense auxiliary signal. On several tasks, approximation error in a student landscape is amplified when it is applied at every state, whereas exact queried values retain the correct local action ranking.”

Results to use. At $\beta = 0.75$, exact queries have higher recorded success during training than uniform distillation in RedBlueDoors, Taxi, and UnlockPickup. The clearest example is RedBlueDoors: the uniform student variant has much lower final success and learning progress than the exact queried variant. DoorKey is the important exception, where distillation is competitive or better; report this plainly.

Suggested figure/table. Use one compact table with Query, Uniform-distill, and the difference for the pre-specified E2 environments. Keep the complete beta sweep in the appendix. Do not add active or uncertainty variants to this figure.

5.3 Action selectivity is the operative ingredient

What to write. Tie the perturbation experiment back to the method: “The centering in Equation (3) is not cosmetic. Matched perturbations show that state-dependent offsets do not produce the observed behavior change; action alignment is required.” Include the $\beta = 0$ identity/null and, if available, an action-permuted counterfactual landscape control. This is the section that makes the paper more than an empirical oracle heuristic.

5.4 Cost and scope

What to write. Report counterfactual branches, wall-clock time, and query fraction for PPO, direct PPO-CF, and PPO-SCA. Do not imply that the oracle is free. State that the present study assumes a simulator state-restoration interface and deterministic/local counterfactual rollouts. This makes the contribution an algorithmic result for structured simulators, not a claim about arbitrary model-free environments.

6 Discussion and limitations

Main discussion. The results suggest that the useful resource is not a generic dense learned bonus. It is a correctly centered, action-level signal at states where an exact counterfactual can be trusted. Explain that this produces a practical spectrum: ordinary PPO uses no simulator intervention; direct PPO-CF uses more complete counterfactual supervision; PPO-SCA obtains much of the benefit from sparse queries.

Limitations to state explicitly.

1. The current evidence is MiniGrid-heavy; a final submission should add at least one non-MiniGrid environment with a valid restoration oracle.
2. The oracle is privileged simulator access. This is not a model-free method unless a learned world model replaces the oracle.
3. The counterfactual horizon and β are task-sensitive.
4. Distillation is not yet a solved scaling mechanism; the reported negative result is part of the conclusion.

Secondary-goal work. Mention only as a future direction or appendix demonstration: an actor-only, reward-free advantage term can induce blue-token collection in a custom DetourPickup environment while retaining goal completion in the available two-seed experiment. Do *not* claim a robust primary–secondary trade-off or general controllability from this result.

7 Appendix plan

1. Full five-environment Notebook 06 curves, including KeyCorridor S3R2.
2. Full E2 beta sweep for exact queried and uniform distilled landscapes.
3. E3 active/uncertainty results, marked as three-seed exploratory results.
4. Detailed derivation of the centered residual and the state-only null.
5. Oracle restoration checks, centering diagnostics, query counts, and computational cost.
6. All configurations, seeds, evaluation seeds, and selection rules.
7. Secondary-goal implementation and its two-seed qualitative result.

Final checklist before converting this pitch into a submission

1. Recreate every headline number with one locked evaluation script.
2. Freeze the environment and β selection rules; run confirmation seeds if the four-task figure remains post-selected.
3. Add Taxi to the common evaluation/plot pipeline or omit it consistently.
4. Add a matched action-permutation control if it is not already complete.
5. Measure query/oracle wall-clock cost against direct PPO-CF.
6. Add one non-MiniGrid task before treating the work as ICLR-ready.